Mini Paper — 1.2 Software & Software Development

Time: ~35 minutes · Total: 30 marks · Answer all questions.

Mark your work against the mark scheme at the end; aim for ~1 minute per mark.

Questions

Q1 (AO1, 2 marks). State what is meant by *virtual memory* and give one reason an operating system uses it.

Q2 (AO1, 3 marks). Describe the difference between **paging** and **segmentation**, and name the type of fragmentation each typically suffers from.

Q3 (AO1/AO2, 4 marks). A multitasking operating system handles an interrupt while a program is running. Explain how the processor handles the interrupt, referring to the **interrupt service routine** and the use of the **stack**.

Q4 (AO2, 4 marks). A games console runs application software and a smart thermostat runs an embedded operating system. (a) State the difference between **application software** and **utility software**, giving one example of each. [2]

(b) Identify the most suitable type of operating system for the smart thermostat and justify your choice. [2]

Q5 (AO1, 4 marks). Describe the four stages of compilation, in order.

Q6 (AO2, 4 marks). Memory contains: address 40 holds the value 25; address 25 holds the value 9. The instruction is `LDA 40`. State the value loaded into the accumulator under (a) immediate, (b) direct and (c) indirect addressing, and explain the indirect case.

Q7 (AO1/AO2, 3 marks). Define **encapsulation** in object-oriented programming and explain one way it makes a program more maintainable or robust.

Q8 (AO3, 6 marks). A small start-up is building a new mobile fitness app. The exact features are still uncertain, the founders want to release something usable quickly and to see and respond to progress regularly, and the budget is limited. Recommend a software development methodology, justify your choice with reference to the scenario, and evaluate it against **at least one** alternative.